

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-2

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The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate. Task to be Done 1.

The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed. 2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. o Hint: Use COUNT(book_id) and GROUP BY published_year. 3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table. 4.

Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title. Skill Set Gained • Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans). •

Analytical SQL: Using GROUP BY to generate business insights.

• Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

STEP1: CREATING THE DATABASE:

```
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\rohit>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 25
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| bookflow_db |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> CREATE DATABASE library_db;
Query OK, 1 row affected (0.01 sec)
```

STEP2: CREATING THE TABLE Books:

CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100)
NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK
(published_year < 2027));

```
mysql> USE library_db;
Database changed
mysql> CREATE TABLE Books(
    -> book_id INT PRIMARY KEY,
    -> title VARCHAR(100) NOT NULL,
    -> isbn VARCHAR(20) UNIQUE,
    -> published_year INT CHECK(published_year <2027)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

STEP3: INSERT BOOKS AND DISPLAY:

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
(1, 'The Great Gatsby', '9780743273565', 1925),  
(2, 'To Kill a Mockingbird', '9780061120084', 1960),  
(3, '1984', '9780451524935', 1949);  
  
SELECT * FROM Books;
```

```
mysql> INSERT INTO Books(book_id,title,isbn,published_year)VALUES  
-> (1, 'The Great Gatsby', '9780743273565', 1925),  
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),  
-> (3, '1984', '9780451524935', 1949);  
Query OK, 3 rows affected (0.09 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Books;  
+-----+-----+-----+-----+  
| book_id | title                | isbn          | published_year |  
+-----+-----+-----+-----+  
| 1       | The Great Gatsby     | 9780743273565 | 1925           |  
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |  
| 3       | 1984                 | 9780451524935 | 1949           |  
+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

STEP4: Members table: Create, Insert and Display:

```
mysql> CREATE TABLE Members(
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Members(member_id,full_name,email)VALUES
  -> (101,'John Smith','john.smith@email.com'),
  -> (102,'Emma Wilson','emma.wilson@email.com'),
  -> (103,'Michael Brown','michael.brown@email.com');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

STEP5: CREATE, Insert, Display: Loans Table (FOREIGN KEYS):

```
mysql> CREATE TABLE Loans(
-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY(member_id) REFERENCES Members(member_id),
-> FOREIGN KEY(book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

Step6: JOIN Query (Join who Borrowed what);

10 rows in set (0.00 sec)

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m ON l.member_id = m.member_id  
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.02 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books  
-> FROM Books  
-> GROUP BY published_year  
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.02 sec)

STEP 7: Donation via Transaction (Atomcity): and Index on ISBN + FAST SEARCH:

```
mysql> CREATE TABLE Donation_History (  
->     donation_id INT PRIMARY KEY,  
->     book_id INT,  
->     donor_name VARCHAR(100),  
->     donation_date DATE,  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.08 sec)  
  
mysql>  
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)  
-> VALUES (1, 4, 'Raj Kumar', CURDATE());  
Query OK, 1 row affected (0.02 sec)  
  
mysql> COMMIT;  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);  
Query OK, 0 rows affected (0.09 sec)  
Records: 0  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';  
+-----+-----+-----+-----+  
| book_id | title | isbn          | published_year |  
+-----+-----+-----+-----+  
|      3 | 1984  | 9780451524935 |      1949      |  
+-----+-----+-----+-----+  
1 row in set (0.01 sec)  
  
mysql> |
```